Van Phan

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Education

Georgia State University *Atlanta, GA*

B.S. Mathematics, GPA: 4.11/4.30 Expected Graduation: May. 2027

- *Relevant Coursework*: Linear Algebra, Differential Equations, Numerical Analysis, Analysis, Modern Algebra, Optimization, Calculus, Biostatistics, Probability and Statistics, Applied Combinatorics, Computer Science

Research Experience

Density and BI-RADS - Aware Triage and Automatic Report Generation for Mammography

Advisor: Dr. Russell Jeter

- Developed a framework to automate diagnostic text generation and enable risk-based case prioritization, flagging scans with extremely dense breasts (ACR Category D) for supplemental screening.
- Fine-tuned the MedGemma 1.5 4B vision-language model using QLoRA on the DMID and VinDr-Mammo dataset, achieving a ROUGE-L score of 0.8012 and a METEOR score of 0.8358 for clinical report generation.
- Accelerated clinical review in simulations on a 100-case cohort by introducing a Cumulative Urgency Score formula, capturing 100.0% of high-risk malignancies, which was only 19.2% in First-In-First-Out workflow, within the first 20% of the reading workload and shifting the mean rank of severe cases from 42.0 down to 7.5.

Time Series Classification of Active Range of Motion in Robotics-Assisted Stroke Rehabilitation

Advisor: Dr. Russell Jeter

- Developed supervised machine learning models to classify range of motion (ROM) in stroke rehabilitation, leveraging 28M+ high-resolution time-series samples from 431 users collected through robotics-assisted therapy.
- Evaluated 11 ML architectures, including XGBoost, LSTM, 1D-CNN, Transformer, InceptionTime, Temporal Convolutional Networks (TCN), and TSFresh-based models, achieving over 43% accuracy on a standardized 100,000-window evaluation set.
- Improved movement-pattern classification by engineering time-series features from Angle and Pressure sensor data, with TSFresh-based embeddings achieving 1.58-bin MAE and LSTM reaching 0.670 AUROC

Morphological Feature Analysis of Retinal Pigment Epithelial Cell from Mice during Aging

Advisor: Dr. Yi Jiang

- Developed an automated end-to-end machine learning pipeline for classifying Retinal Pigment Epithelial (RPE) cell during aging, generating insights for potential therapeutic strategies in Age-related Macular Degeneration.
- Achieved highly accurate classification as measured by 90%+ cross-validation F1-score, by extracting 133 morphological and texture-based features from 326 RPE cell images and implementing a stacking ensemble of XGBoost, LightGBM, and CatBoost with a Logistic Regression meta-learner.
- Identified critical RPE aging biomarkers, suggesting Local Binary Patterns (LBP) and geometric shape metrics (circularity, aspect ratio) as the most influential classification variables.

Modeling Wound Healing Without Migration and Proliferation

Advisor: Dr. Yi Jiang

- Developed a mathematical framework to model Retinal Pigment Epithelium (RPE) wound healing, supporting the advancement of Age-related Macular Degeneration (AMD) therapeutics.
- Simulated purse-string contraction and cell fusion repair mechanisms in Python-based CompuCell3D, by utilizing a Q-Potts cellular model and Monte Carlo evaluation of the Hamiltonian free energy states.
- Quantified wound closure efficiency across algorithmic iterations by analyzing texture tensors and spatial points, proving that lower cellular volume constraints, higher wound's radius and affected area yield statistically slower wound closure.

Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion

Advisor: Dr. Yi Jiang

- Engineered a Physics-Informed Neural Networks (PINNs) model to simulate collective cancer invasion, resulting in a data-efficient, mesh-free solution for complex Partial Differential Equations (PDEs).
- Developed a framework utilizing Tensorized Fourier Neural Operators (TFNO) and PyTorch's autograd engine to encode physical laws (PDEs, boundary, and initial conditions) directly into the model's loss function, reducing reliance on labeled data.

- Explored advanced techniques, including a sequence-to-sequence (Seq2seq) PINO approach and Augmented Lagrangian methods (ALM), to overcome model failure modes like vanishing gradients and improve solution accuracy and stability.

Laboratory Rearing Model of *Lissachatina fulica* for Acharan Sulfate Extraction & Evaluation

- Established a laboratory rearing model of *Lissachatina fulica*, using garlic essential oil and *Lactobacillus casei* for inhibition of pathogenic bacteria (*E. coli*, *S. aureus*, and *Salmonella spp.*).
- Extracted acharan sulfate of snail flesh and mucus from *Lissachatina fulica*.
- Investigated anticoagulant and thrombolytic properties of extracted acharan sulfate, achieving 89.40% anticoagulant activity within 30 minutes via clotting time assay and 32.09% thrombolytic activity within 1 hour via clot lysis assay.

Development of a Miniaturized Patient Lift System Using 3D Printing and Stepper Motor Control

- Developed a miniaturized model of a patient lift machine using 3D Printing, Arduino A4988 Stepper Motor, and Arduino Uno REV3 ICSP microcontroller board.
- Enhanced lifting and moving capabilities as compared to existing models, while simultaneously reducing the size.

Publications

- **Van Phan**, Nguyen Nhat Cuong Tran, Ngo Tan Dat Bui, and Russell Jeter. "DB-ATRG: The Density and BI-RADS-Aware Triage and Automatic Report Generation System for Mammography" [In Review] (2026). doi: 10.64898/2026.07.22.26358655.

Posters & Presentations

Density and BI-RADS - Aware Triage and Automatic Report Generation System for Mammography,

- Poster presented at **SIAM Conference on Mathematics of Data Science**, 2026.
- Poster presented at Georgia State Undergraduate Research Conference, 2026.

Time Series Classification of Active Range of Motion in Robotics-Assisted Stroke Rehabilitation,

- Poster presented at **SIAM Conference on Mathematics of Data Science**, 2026.

Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion,

- **Best Project Award** at Summer Undergraduate Research Symposium, 2025.

Morphological Feature Analysis of Retinal Pigment Epithelial Cell from Mice during Aging,

- Poster presented at Georgia State Undergraduate Research Conference, 2025.

Modeling Wound Healing Without Migration and Proliferation,

- Poster presented at Summer Undergraduate Research Symposium, 2024.

Laboratory Rearing Model of *Lissachatina fulica* for Acharan Sulfate Extraction & Evaluation,

- Oral presentation at Vietnam Science and Engineering Fair (ViSEF), 2022.

Work Experience

Viet Home Care LLC Jun. 2024 – Sep. 2026
Data Science Intern *Atlanta, GA*

- Developed data pipelines integrating Vision-Language Models (VLMs) to extract and structure information from health records, health coaching sessions, and home health visits, transforming unstructured healthcare data into analysis- and model-ready datasets.
- Built machine learning models to optimize personal care aide matching based on client needs and compatibility across a pool of 1,000+ clients and aides.
- Analyzed service utilization and operational data across several services, generating actionable insights to improve service processes, quality, and outcomes for 1,000+ households.

Georgia State University Aug. 2025 – Aug. 2026
Chemistry Laboratory Teaching Assistant *Atlanta, GA*

- Mentored 50+ undergraduate students in Survey of Chemistry I and II, improving understanding of fundamental principles and reducing procedural errors through one-on-one guidance and demonstration of proper laboratory techniques.

Center for the Advancement of Students and Alumni Jun. 2025 – Aug. 2025
Near-peer Mentor *Atlanta, GA*

- Provided near-peer mentorship and research support to students in the Math Path Program, guiding research technique implementation and project development for the Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion project.

Honors & Awards

- **Best Project Award** at Summer Undergraduate Research Symposium (2025)
- **President's List** at Georgia State University (2023, 2024, 2025, 2026)
- **Campus Atlanta Scholarship** at Georgia State University (2023 – 2027)
- **Prospect Award (National Level) & First Place (City Level)** at Vietnam Science and Engineering Fair (ViSEF) (2022)
- **Consolation Prize** at Engineering and Science Fair of Ho Chi Minh City University of Technology (2022)
- **Bronze Prize** in Physics at the Ho Chi Minh City Excellent Student Contest (2019)

Technical Skills

- **Programming & Tools**: Python, Julia, R, MATLAB, SQL, Git/GitHub, Linux/CLI, LaTeX, CompuCell3D
- **Scientific & Math Modeling**: Physics-Informed Neural Networks, PDE Modeling, Tensorized Fourier Neural Operators, Augmented Lagrangian Method, Q-Potts Cellular Modeling, Monte Carlo Evaluation
- **ML, LLM & AI**: LLMs, Fine-Tuning, Quantization, Transformers, TRL, Hugging Face, Ensemble Methods, Time-Series Classification (TCN, InceptionTime, LSTM, 1D-CNN), Time-Series Embedding
- **Libraries**: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CatBoost, TSFresh, OpenCV, Pandas, NumPy, SciPy, Scikit-image, Pillow, Mahotas, Matplotlib, Seaborn

References

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